STAT536HW2

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```
set.seed(2427348)
library(tidyverse)
library(nnet)
library(MASS)
library(ordinal)
library(broom)
library(janitor)
```

Problem 1

Data Preprocessing

```
mental <- read_table("mentalimpairment-data.txt", col_names = TRUE) %>%
  mutate(
    impairment = factor(Impairment, levels = c("Well", "Mild", "Moderate", "Impaired")),
    impairment_ord = factor(Impairment, ordered = TRUE, levels = c("Well", "Mild", "Moderate", "Impaire
    ses = factor(SES, levels = c(0, 1), labels = c("low", "high")),
    life = Events
  )

str(mental)
```

```
## tibble [40 x 8] (S3: tbl_df/tbl/data.frame)
##  $ Subject       : num [1:40] 1 2 3 4 5 6 7 8 9 10 ...
##  $ Impairment    : chr [1:40] "Well" "Well" "Well" "Well" ...
##  $ SES           : num [1:40] 1 1 1 1 0 1 0 1 1 1 ...
##  $ Events        : num [1:40] 1 9 4 3 2 0 1 3 3 7 ...
##  $ impairment    : Factor w/ 4 levels "Well","Mild",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ impairment_ord: Ord.factor w/ 4 levels "Well"<"Mild"<..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ ses           : Factor w/ 2 levels "low","high": 2 2 2 2 1 2 1 2 2 2 ...
##  $ life          : num [1:40] 1 9 4 3 2 0 1 3 3 7 ...
```

Part A

```
fitA_full <- multinom(impairment ~ ses + life, data = mental, trace = FALSE)
fitA_ses <- multinom(impairment ~ life, data = mental, trace = FALSE)
fitA_life <- multinom(impairment ~ ses, data = mental, trace = FALSE)
```

```r
A_LRT_ses <- anova(fitA_ses, fitA_full, test = "Chisq")
A_LRT_life <- anova(fitA_life, fitA_full, test = "Chisq")
A_LRT_ses; A_LRT_life
```

```
## Likelihood ratio tests of Multinomial Models
##
## Response: impairment
##        Model Resid. df Resid. Dev   Test    Df LR stat.   Pr(Chi)
## 1       life       114  101.81640
## 2 ses + life       111   96.69826 1 vs 2     3 5.118134 0.1633484

## Likelihood ratio tests of Multinomial Models
##
## Response: impairment
##        Model Resid. df Resid. Dev   Test    Df LR stat.    Pr(Chi)
## 1        ses       114  105.28471
## 2 ses + life       111   96.69826 1 vs 2     3 8.586448 0.03532589
```

```r
fitA_best <- fitA_full
if (A_LRT_ses$`Pr(Chi)`[2] > 0.05) fitA_best <- fitA_ses
if (A_LRT_life$`Pr(Chi)`[2] > 0.05 && A_LRT_ses$`Pr(Chi)`[2] <= 0.05) fitA_best <- fitA_life

summary(fitA_best)
```

```
## Call:
## multinom(formula = impairment ~ life, data = mental, trace = FALSE)
##
## Coefficients:
##          (Intercept)      life
## Mild      -0.7801769 0.2173599
## Moderate  -1.2512536 0.2011724
## Impaired  -2.4536330 0.4847313
##
## Std. Errors:
##          (Intercept)      life
## Mild       0.7559415 0.1780997
## Moderate   0.8809727 0.2019959
## Impaired   1.0490171 0.2014681
##
## Residual Deviance: 101.8164
## AIC: 113.8164
```

```r
johnA <- tibble(ses = factor("low", levels = levels(mental$ses)), life = 7)
predA <- predict(fitA_best, newdata = johnA, type = "probs")
round(predA, 4)
```

```
##     Well     Mild Moderate Impaired
##   0.1465   0.3074   0.1714   0.3748
```

The likelihood ratio tests indicate that life events significantly influence mental impairment levels, while economic status does not. Therefore, the final model includes only the number of life events as a predictor.

The results suggest that individuals experiencing more important life events are more likely to suffer from higher levels of mental impairment. John's most probable mental impairment level is Impaired, indicating that frequent stressful life events substantially increase the likelihood of mental health problems.

Part B

```r
fitB_full <- clm(impairment_ord ~ ses + life, data = mental, link = "logit")
fitB_ses <- clm(impairment_ord ~ life, data = mental, link = "logit")
fitB_life <- clm(impairment_ord ~ ses, data = mental, link = "logit")

B_LRT_ses <- anova(fitB_ses,  fitB_full)
B_LRT_life <- anova(fitB_life, fitB_full)

get_p <- function(tab) {
  nm <- names(tab)
  pcol <- nm[grepl("^Pr", nm)]
  if (length(pcol) == 0) return(NA_real_)
  as.numeric(tab[[pcol]][nrow(tab)])
}

p_ses <- get_p(B_LRT_ses)
p_life <- get_p(B_LRT_life)

fitB_best <- fitB_full
if (!is.na(p_ses) && p_ses  > 0.05) fitB_best <- fitB_ses
if (!is.na(p_life) && p_life > 0.05 && !( !is.na(p_ses) && p_ses > 0.05 )) fitB_best <- fitB_life

summary(fitB_best)
```

```
## formula: impairment_ord ~ life
## data:    mental
##
##  link  threshold nobs logLik AIC    niter max.grad cond.H
##  logit flexible  40   -51.26 110.53 4(0)  2.77e-09 3.1e+02
##
## Coefficients:
##      Estimate Std. Error z value Pr(>|z|)
## life   0.2879     0.1175   2.451   0.0142 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Threshold coefficients:
##                   Estimate Std. Error z value
## Well|Mild           0.2614     0.5639   0.464
## Mild|Moderate       1.6563     0.6171   2.684
## Moderate|Impaired   2.5876     0.6938   3.730
```

```r
johnB <- tibble(ses = factor("low", levels = levels(mental$ses)), life = 7)
predB <- predict(fitB_best, newdata = johnB, type = "prob")
probB <- predB$fit %>% as_tibble()
round(probB, 4)
```

```
## # A tibble: 1 x 4
##    Well  Mild Moderate Impaired
##   <dbl> <dbl>    <dbl>    <dbl>
## 1 0.148 0.264    0.228    0.361
```

The ordinal logistic regression model indicates that life events have a statistically significant positive effect on mental impairment, while socioeconomic status is not significant. Thus, the final model includes only life events as a predictor. This positive coefficient means that as the number of life events increases, the probability of being in a higher impairment category also increases. John's most likely category is Impaired, indicating that frequent life events are strongly associated with greater mental impairment severity.

Problem 2

Data Preprocessing

```r
allig_raw <- read_table("alligatorfood-data.txt", col_names = TRUE)


allig <- allig_raw %>%
pivot_longer(cols = Fish:Other, names_to = "Food", values_to = "Count") %>%
mutate(
lake = factor(Lake),
gender = factor(Gender),
size = factor(Size, levels = c(0, 1), labels = c("<=2.3", ">2.3")),
Food = factor(Food)
)

str(allig)
```

```
## tibble [80 x 8] (S3: tbl_df/tbl/data.frame)
##  $ Lake  : chr [1:80] "Hancock" "Hancock" "Hancock" "Hancock" ...
##  $ Gender: chr [1:80] "Male" "Male" "Male" "Male" ...
##  $ Size  : num [1:80] 0 0 0 0 0 1 1 1 1 1 ...
##  $ Food  : Factor w/ 5 levels "Bird","Fish",..: 2 3 5 1 4 2 3 5 1 4 ...
##  $ Count : num [1:80] 7 1 0 0 5 4 0 0 1 2 ...
##  $ lake  : Factor w/ 4 levels "George","Hancock",..: 2 2 2 2 2 2 2 2 2 2 ...
##  $ gender: Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 2 2 2 2 ...
##  $ size  : Factor w/ 2 levels "<=2.3",">2.3": 1 1 1 1 1 2 2 2 2 2 ...
```

```r
fit_full <- multinom(Food ~ lake + gender + size, weights = Count, data = allig, trace = FALSE)
summary(fit_full)
```

```
## Call:
## multinom(formula = Food ~ lake + gender + size, data = allig,
##     weights = Count, trace = FALSE)
##
## Coefficients:
##              (Intercept) lakeHancock lakeOklawaha lakeTrafford  genderMale
## Fish           2.4322304  -0.5751295    0.5513785  -1.23681053  0.60674971
## Invertebrate   2.6012531  -2.3557377    1.4645820  -0.08096493  0.14378459
```

```
## Other          1.0014505   0.1914537    0.5775317   0.32097943  0.35423803
## Reptile       -0.9829064   0.5534169    3.0807416   1.82333205 -0.02053375
##                size>2.3
## Fish         -0.7308535
## Invertebrate -2.0671545
## Other        -1.0214847
## Reptile      -0.1741207
##
## Std. Errors:
##              (Intercept) lakeHancock lakeOklawaha lakeTrafford genderMale
## Fish           0.7706940   0.7952147     1.210229    0.8661187  0.6888904
## Invertebrate   0.7917210   0.9463640     1.232835    0.8814625  0.7292510
## Other          0.8747773   0.9072182     1.374545    0.9589807  0.7623738
## Reptile        1.2827234   1.3797755     1.591542    1.3388017  0.9088217
##               size>2.3
## Fish         0.6523273
## Invertebrate 0.7084028
## Other        0.7250455
## Reptile      0.8555051
##
## Residual Deviance: 537.8655
## AIC: 585.8655
```

```r
exp(coef(fit_full))
```

```
##              (Intercept) lakeHancock lakeOklawaha lakeTrafford genderMale
## Fish          11.3842454  0.56263198     1.735644    0.2903087  1.8344592
## Invertebrate  13.4806195  0.09482353     4.325735    0.9222260  1.1546354
## Other          2.7222276  1.21100878     1.781635    1.3784772  1.4250944
## Reptile        0.3742219  1.73918556    21.774544    6.1924577  0.9796756
##               size>2.3
## Fish         0.4814978
## Invertebrate 0.1265454
## Other        0.3600600
## Reptile      0.8401955
```

```r
fit_lg <- multinom(Food ~ lake + gender, weights = Count, data = allig, trace = FALSE)
anova(fit_lg, fit_full, test = "Chisq")
```

```
## Likelihood ratio tests of Multinomial Models
##
## Response: Food
##                   Model Resid. df Resid. Dev   Test    Df LR stat.     Pr(Chi)
## 1         lake + gender       300   555.4658
## 2 lake + gender + size       296   537.8655 1 vs 2     4 17.60028 0.001476996
```

```r
newdat <- tibble(
  lake = factor("Trafford", levels = levels(allig$lake)),
  gender = factor("Male", levels = levels(allig$gender)),
  size = factor(">2.3", levels = levels(allig$size))
)
```

```r
prob_pred <- predict(fit_full, newdata = newdat, type = "probs")

round(prob_pred, 4)
```

```
##         Bird         Fish Invertebrate        Other      Reptile
##       0.1045       0.3051       0.1898       0.2012       0.1993
```

```r
prob_reptile <- unname(prob_pred["Reptile"])
prob_reptile
```

```
## [1] 0.1993438
```

The multinomial logistic regression model reveals that both lake and size are significant predictors of alligators' primary food choice. Alligators from Lake Trafford show a strong preference for reptiles, while larger alligators are less likely to consume fish or invertebrates. Gender does not appear to have a meaningful effect. According to the model, a male alligator from Lake Trafford that is three meters long has an estimated probability of about 19.9% of preferring reptiles for dinner.